



Article

Microcredit Pricing Model for Microfinance Institutions under Basel III Banking Regulations [†]

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Abstract: Purpose. The purpose of this research is to propose a tool for designing a microcredit risk pricing strategy for borrowers of microfinance institutions (MFIs). Design/methodology/approach. Considering the specific characteristics of microcredit borrowers, we first estimate and measure microcredit risk through the default probability, applying a parametric technique such as logistic regression and a non-parametric technique based on an artificial neural network, looking for the model with the highest predictive power. Secondly, based on the Basel III internal ratings-based (IRB) approach, we use the credit risk measurement for each borrower to design a pricing model that sets microcredit interest rates according to default risk. Findings. The paper demonstrates that the probability of default for each borrower is more accurately adjusted using the artificial neural network. Furthermore, our results suggest that, given a profitability target for the MFI, the microcredit interest rate for clients with a lower level of credit risk should be lower than a standard, fixed rate to achieve the profitability target. Practical implications. This tool allows us, on the one hand, to measure and assess credit risk and minimize default losses in MFIs and, secondly, to promote their competitiveness by reducing interest rates, capital requirements, and credit losses, favoring the financial self-sustainability of these institutions. Social implications. Our findings have the potential to make microfinance institutions fairer and more equitable in their lending practices by providing microcredit with risk-adjusted pricing. Furthermore, our findings can contribute to the design of government policies aimed at promoting the financial and social inclusion of vulnerable people. Originality. The personal characteristics of microcredit clients, mainly reputation and moral solvency, are crucial to the default behavior of microfinance borrowers. These factors should have an impact on the pricing of microcredit.

Keywords: microfinance institutions; credit risk; neural network; logit; pricing



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1. Introduction

The microfinance industry has been part of the most important international development policies in recent decades (Bettoni et al. 2023). Traditionally, microfinance institutions have traditionally provided resources to groups of people excluded from the traditional financial system through small loans as poverty alleviation, as well as boosting women's empowerment in the labor force (Pietrapiana et al. 2021; Rahayu 2020; Karlan and Zinman 2011).

Previous research has considered the creation of microenterprises in the most disadvantaged regions as a way to combat poverty by increasing the innovation and competitiveness of an economy and reducing the unemployment rate (Aguilar and Portilla 2019; Wennekers et al. 2005). However, microentrepreneurs have traditionally experienced great difficulties in accessing savings and credit services due to a lack of collateral and assets with which to raise finance, thus preventing them from prospering through their business projects.

thus restricting the supply of microcredits and, consequently, increasing interest rates. The research suggests that regulators develop specific guidelines for these models and that MFI managers expand their approaches to include historically overlooked areas, such as the agricultural sector and rural communities. This is important for improving access to financial services and supporting local economic development.

Furthermore, our findings can contribute to the design of government policies aimed at promoting the financial and social inclusion of vulnerable people. Identifying risk factors in MFI clients can help governments target subsidies to the most creditworthy borrowers, thereby improving the performance of these institutions and their social contribution.

4.4. Limitations of the Research

While the results of this research offer valuable insights into the implementation of risk-adjusted pricing models in the microfinance sector, it is important to recognize some limitations. First, the study's reliance on historical data from a specific geographic region, such as Guatemala, may restrict the applicability of the results to other contexts. Microfinance environments vary considerably across countries due to differences in regulatory frameworks, economic conditions, and cultural factors. Therefore, the effectiveness of the proposed pricing strategy may not be the same in other regions with different financial scenarios.

In addition, while the use of neural network models to predict default risk is innovative, it introduces challenges in terms of model interpretability and transparency. Stakeholders in the microfinance sector, especially those with limited technical knowledge, may find it difficult to understand and trust the decisions generated by these models. This could hinder the widespread adoption of these advanced techniques, especially in contexts where regulators require high levels of model transparency.

In addition, the data used in this study, while comprehensive, may not fully reflect all the nuances of borrower behavior and external economic shocks that may affect default rates. For example, unexpected macroeconomic events or changes in local economic conditions were not included in the model, which could affect its predictive accuracy.

Finally, it is critical to consider the ethical implications of applying risk-based pricing within microfinance. While this approach could lead to fairer interest rates, there is a risk of inadvertently excluding the riskiest borrowers, who are often the most vulnerable. This raises significant questions about the balance between financial sustainability and social mission in microfinance, a tension that future research should explore further.

5. Conclusions

In keeping with their important social mission, MFIs must improve their management models if they want to survive in an unstable macroeconomic environment and in a microfinance sector in which commercial banks have been gaining a foothold in recent years. To achieve this goal, with criteria of effectiveness and efficiency, MFIs can gain competitiveness with respect to commercial banks through tools that allow them to calculate probabilities of default in a reliable and robust way, which would make it possible to calculate capital requirements and set interest rates adjusted to the credit risk of each client under the IRB approach of the Basel III regulations.

Our findings represent an advance on the conclusions of previous research for three reasons. First, building on previous research, our results support that a probability of default model using a multilayer-perceptron model offers better accuracy returns associated with lower classification costs than the classical parametric technique of logistic regression. Moreover, our results imply important implications for improving the efficiency, solvency, and profitability of MFIs due to their contribution to the reduction in default losses, bankruptcy provisioning, and recovery processes. In this respect, our results suggest that multilayer perceptron neural networks are the most suitable techniques in the microfinance sector.

Second, our research corroborates that capital requirements under the IRB approach are quantified much more accurately than under the old standard method. Moreover, through this approach, MFIs will be able to improve the profitability and efficiency of their management, which would also increase their ability to raise funding in the financial markets under more competitive conditions.

Third, the main finding of our research is that a pricing model in the microfinance industry is an important tool that allows MFIs to defend and increase their market share in order to continue carrying out their original mission, which is to finance the poor without the need to charge high interest rates. By randomly selecting three clients with different risk profiles, our results establish a priori the desired profitability for each of them, allowing the institution to maximize its profits and meet the demands of its shareholders and funders. Consequently, these models are seen as management tools that can help to increase efficiency and reduce microcredit default losses so that MFIs can achieve the necessary financial self-sufficiency and compete on a level playing field with traditional banks. Furthermore, our findings can contribute to the design of government policies aimed at promoting the financial and social inclusion of vulnerable people. Moreover, by increasing solvency and profitability, MFIs ensure their survival in a sector where commercial banks have entered through downscaling.

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